Fundamentals of ICT

1.INTRODUCTION TO ICT

- ICT stands for Information and Communication Technology which encompasses technologies for handling communication, information, audio processing and monitoring functions and more.
- Information and communication technologies (ICT) is defined as **a diverse set of technological tools and resources used to create, share, store and transmit or exchange information**.
  - These technological tools and resources include computers, the Internet (websites, blogs and emails), live broadcasting technologies (radio, television and webcasting), recorded broadcasting technologies (podcasting, audio and video players, and storage devices) and telephony (fixed or mobile, satellite, visio/video-conferencing, etc.).

DIFFERENCE BETWEEN IT,COMPUTER SCIENCE AND ICT

- IT(Information Technology) focuses on the practical application of computer systems and networks for business operations
- Computer Science concentrates on the theoretical and algorithmic aspects of computing and covers a broader range of topics
  - Computer science focuses on the development and testing of software and software systems.
  - It involves working with mathematical models, data analysis and security, algorithms, and computational theory.
  - Computer scientists define the computational principles that are the basis of all software.
- Information and Communication Technology involves the use of technology for communication ,information processing and other interconnected applications

- IT (information technology) encompasses the use of computers, networks, computer software and other electronic or digital devices for the management and communication of information.
- ICT (information communication technology) focuses more on how digital technologies assist users in handling information.

USES OF ICT

- Digital era
  - ICT has made it easier for people to communicate with each other through its technology.
  - Instead of face-to-face time-consuming conversations, nowadays it is easier to pass on information through calls, texts, and emails.
  - **Digital** refers to electronic technology that uses discrete values, generally zero and one, to generate, store and process data.
  - In **digital technology**, data is transmitted and stored as strings of zeros and ones, each of which are referred to as bits.
- Career opportunities
  - Programmers, Database administrators, Network Enginners, Computer Service Technicians, Data Analysts, Web developers, Systems analysts, IT Consultants, Software Analyst, System Administrator, IT Managers, Hardware engineers, Cybersecurity specialist, Software developers etc
- Innovation
  - ICT - influence growth of economies and businesses efficiency. It ha
  - Artificial Intelligence

2.INTRODUCTION TO COMPUTERS

- A computer is an electronic device that processes data, performs calculations and executes instructions to perform various tasks.
- An electronic device that processes data according to a set of stored instructions.
- An electronic devices that can follow instructions **to** accept input, process the input and then produce **information**.
- It is a machine that manipulates data according to a set of instructions.

EXAMPLES OF COMPUTERS

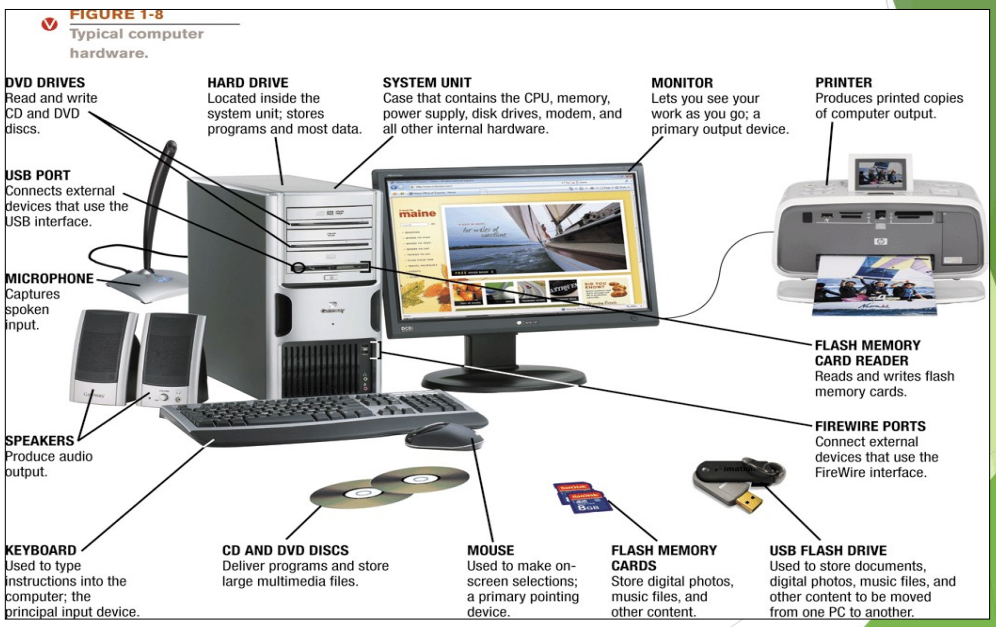
- Personal computers
- Servers
- Mainframes
- Gaming consoles

Computers are made of

- Hardware
  - Physical parts of a computer
- Software
  - A general term for programs

Hardware

- Hardware:
  - The physical parts of a computer
  - Refers to the physical components of a computer system that can one can see, touch and manipulate
    - Internal hardware
      - Located inside the main box (system unit) of the computer i.e. Hard disk,
    - External hardware
      - Located outside the system unit and plug into ports located on the exterior of the system unit e.g. keyboard, mouse, screen/monitor etc



Software

- ***Software*** is a general term for different kinds of *programs* used to operate a computer and its related devices.
- ***Software*** are the programs that run on a computer and perform certain functions
- **Two types of Software:**
  - *System software*
    - A computer software designed to operate and control computer hardware and to provide a platform for running application software.
    - Example of system software is operating system which allow a computer to operate e.g. Windows 7, 8, 10, Linux etc.
    - Boots the computer and launches programs at the user's direction
  - *Application software:*
    - Programs that perform specific tasks or applications for the user
    - A program or group of programs designed for end users.

Computer virus

- A damaging program that affects or infects a computer negatively by altering the way the computer works without the user's knowledge or permission.
- Computer viruses are small programs that are designed to spread from one computer to another and interfere with computer operation.
- A virus might corrupt or delete data on a computer.

- A virus
  - Infects programs, documents, databases and more
  - It is man-made
  - It can hide and reproduce
  - It can lay dormant (inactive) and then activate

- **Three main sources of viruses:**
  - Internet
    - Via downloads and exchanges
  - USB flash disks
    - Exchanging disks
  - Computer networks
    - Can spread from one computer to another on a network.

Virus Protection

- The best way to cope with viruses is to recognize their existence and use an **Antivirus software**.

Data

- These are raw facts or observations that have little or no value until they have been processed and transformed into information.
  - E.g. date, price, weight, age etc

Information

- Is data that has been processed so that they are meaningful and is useful in solving problems.
  - E.g. bank statement

Information Technology

- Technology
  - Application of scientific knowledge for practical purposes.
- Information Technology
  - Any technology concerned with the capture, storage, transmittal or presentation of information.

Information System (IS)

- Any system that make use of IT
- The role of information system is to provide information to senior managers which will enable them to make decisions to ensure that the organisation is controlled.
- IS affect:

  - how manager decide

  - how senior manager plan

  - what produce and service are produced and how

INFORMATION SYSTEMS

- Information systems refer to organized, purposeful systems designed to collect, process, store and distribute information.

TYPES OF INFORMATION SYSTEMS

- Management information systems provides middle managers with reports and information to support decision making.
- Transaction processing systems manages and records daily transactions such as sales or purchases e.g. ATM (Automated Teller Machines, Order processing systems, Airlines seat reservation systems, student records system, manufacturing, shipping etc
- Decision support system assists in decision making by providing interactive , real time information and analysis tools e.g. Geographical Information system, credit loan verification systems, medical diagnosis systems etc
- Expert systems utilizes artificial intelligence to mimic human expertise in a specific domain
  - A computer program that simulates the judgment of a human expert is known as an expert system in ai. A few examples of an expert system are DENDRAL, a molecular structure prediction tool for chemical analysis. Another example of an expert system that predicts the kind and extent of lung cancer is PXDES.
- Executive information systems offers high level executives summarized reports and access to critical information

The Role Of Computers In ICT

- Teleworking or Telecommuting
  - Making use of networking technology to allow working at different locations while keeping in touch with co-workers.
- Electronic Trading
  - Organisations and their customers making use of networks as a means for advertising, buying and selling goods and services.
- Tele and video conferencing
  - Groups of people conducting meetings via a telephone or video network.

- Computers in education
  - Use *presentation software* such as PowerPoint to show lecture outlines and other materials on classroom screens.
  - *Distance learning*, or *e-learning* - Instead of a book it is easier to learn from e – learning software.
- Computers in business
  - Makes businesses more profitable
  - ecommerce
- Computers in health
  - Maintain and access patient records
  - *Telemedicine* —medical care delivered via telecommunications.
  - *Telemedicine*, health-care professionals in separate locations conduct live conferences on the computer.
  - "Teleradiology" to exchange computerized images such as X rays via telephone-linked networks
- Computers are used to design products
  - Assembly lines are automated

- Computers at home
  - Business - ecommerce
  - Entertainment – music and movies
  - Communication – emails, chats
  - Education - eLearning
- Computers in government
  - Necessary to track data for population
  - Police officers
  - Tax calculation and collection
  - Governments were the first computer users

- Money:
  - Toward the Cashless Society
  - *ATM Machines* – Software authenticate users and dispense cash.
- Leisure:
  - Computers in entertainment and arts
  - *Digital video and audio composition* – Editing and composition of audio and video have been made much easier by a computer.
- Ways For You To Find Employers

Advantages of using computers

- **Speed**
  - Computers can process millions of instructions each second in a very short time
- **Accuracy**
  - Calculations done by a computer is likely to be completely accurate.
  - Human errors such as typing errors can be reduced or eliminated.
- **Reliability**
  - Computer based systems are dependable and reliable because they rarely break or fail.
- **Consistency:**
  - Given the same input and processes, a computer will produce the same results — consistently.

- **Storage:**
  - A computer can transfer data quickly from storage to memory, process it, and then store it again for future use.
  - Computers keep large amount of data and make this data available for processing anytime it is needed.
- **Communications:**
  - Computers today can communicate with other computers, often wirelessly.
  - Computers with this capability can share any of the four information processing cycle operations — input, process, output, and storage — with another computer or a user.
  - Can send and retrieve data from other computers if in a network

Computers greatly speed up problem solving and other tasks thereby increasing users' productivity.

Disadvantages of using computers

- **Health Risks:**
  - Prolonged or improper computer use can lead to injuries or disorders of the hands, wrists, elbows, eyes, neck and back.
- **Violation of Privacy:**
  - Almost every life event is kept in a computer somewhere . . .such as medical records, payroll, credit reports, tax records, etc.
  - Sometimes personal and confidential records are not protected properly,
    - individuals have found their privacy violated and identities stolen.
- **Public Safety:**
  - People around the world are using computers to share publicly their photos, videos, journals, music and other personal information.
  - Some innocent computer users have fallen victim to crimes committed by dangerous strangers.
  - *Protect yourself and your dependents from these criminals by being cautious in e-mail messages and on Web sites.*

- **Impact on Labor Force:**
  - The skills of millions of employees have been replaced by computers.
  - Therefore, it is essential that workers keep their education up-to-date.
  - Some companies are outsourcing jobs to foreign countries instead of keeping their homeland labour force employed.
- **Impact on Environment:**
  - Computer manufacturing processes and computer waste are depleting natural resources and polluting the environment.
  - When computers are thrown away in landfills, they can release toxic materials and potentially dangerous levels of lead mercury and flame retardants.

Disadvantages of using computers

- Require skilled manpower for design and programming and data encoding.
- Require electricity.
- Require air conditioning for non-stop work.
- Expensive to acquire and maintain
- Require frequent upgrade due to fast developments in hardware and software.
- Require regular staff training.